

PrivEdit: A Zero-shot Interactive Image Privacy Editing System

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Abstract

In this paper, we present PrivEdit, a zero-shot, interactive image privacy editing system specifically designed for automated sensitive information desensitization. As social networks and smart devices proliferate, the risk of unintended privacy leakage grows, driving demand for personalized, controllable protection tools. PrivEdit is powered by natural-language instructions and integrates a Recognize-Anything model for robust detection and classification of sensitive objects (e.g., faces, license plates, ID cards), followed by GroundingDINO and SAM for high-precision mask extraction. User intents are parsed and disambiguated via GPT-4o, enabling selective target confirmation and iterative refinement. Finally, our editing module performs localized edits—such as adjustable blurring, mosaicking, or replacement via generative editing. With support for multi-round feedback and real-time modification, PrivEdit seamlessly handles both pre-recorded images and live streams, making it ideal for social-media pre-publishing, privacy data desensitization in enterprise or healthcare contexts, and intelligent surveillance applications. By unifying detection, segmentation, intent parsing, and localized editing into one coherent interface, PrivEdit delivers an end-to-end solution for safeguarding visual data. Supplementary materials including the demo video and slides are available at: [Google Drive folder](#).

CCS Concepts

• Security and privacy → Privacy protections.

Keywords

Privacy Protection, Local Image Editing, Interactivity

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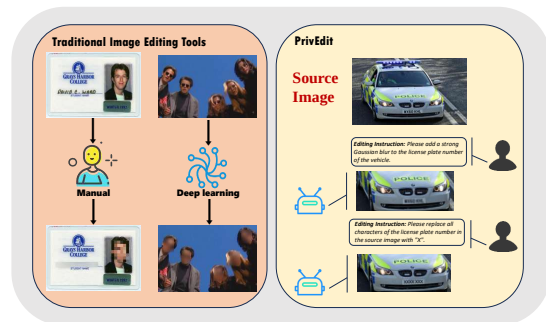


Figure 1: Comparison with Traditional Image Editing Tools. Unlike traditional manual or deep learning-based image editing approaches, PrivEdit allows users to interactively control the editing process via natural language instructions, ensuring privacy protection through automatic segmentation and editing of sensitive information in images.

1 Introduction

In the digital media era, capturing and sharing photos has become easier, but it also raises significant concerns about privacy leaks. Images often contain sensitive information, such as faces, license plates, or IDs, which can pose serious privacy risks if shared unprotected. Traditional privacy protection methods rely on manual or basic tools like Photoshop[1] and GIMP[2], which require users to manually select regions and adjust parameters, offering limited flexibility. Moreover, many deep-learning-based solutions offer one-shot editing with fixed parameters (e.g., “blur all faces”) and lack fine-grained control or iterative refinement, making them less adaptable to varied needs.

To address these limitations, this paper proposes the PrivEdit, a system which enables zero-shot, interactive editing of sensitive areas in images through natural-language directives. Users can simply enter commands such as “blur all faces in the photo” or

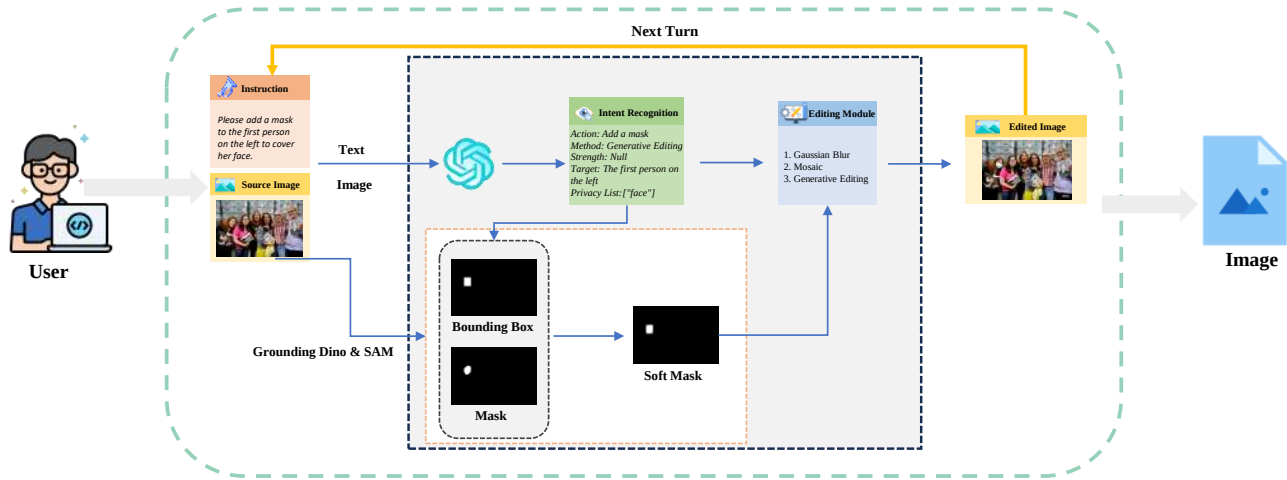


Figure 2: The overview of PrivEdit case mainly follows four steps: Intent Recognition, where user input is analyzed to identify the editing action; Automatic Segmentation, using Grounding DINO & SAM to detect sensitive areas; Editing, applying the chosen edits (e.g., Gaussian Blur, Mosaic); and Multi-Round Iterative Feedback, allowing users to refine the edits through multiple iterations until satisfied.

“apply a mosaic to the license plate,” and the system will automatically identify the specified targets and generate the required edits. PrivEdit integrates multiple state-of-the-art models: it leverages a large-language model[3] to parse instructions and confirm editing objectives; employs the Recognize-Anything[4–6] framework for general object detection; uses Grounded SAM[7], an integrated pipeline that combines GroundingDINO[8] and Segment Anything Model (SAM)[9, 10] for precise segmentation; and finally applies an integrated local editing module (which incorporates adjustable Gaussian blur, mosaicking, and generative editing via the ACE++[11] framework) to perform high-quality modifications on the designated regions. The system supports adjustable blur intensities, target-region selection, and multi-turn interactive feedback, thereby granting users fine-grained control over the privacy-protection process.

2 PrivEdit

The architecture of PrivEdit, as shown in Figure 2, includes four steps: Intent Recognition, Automatic Segmentation, Editing and Multi-Round Iterative Feedback.

Intent Recognition:[12] Upon receiving a natural language instruction from the user, a Large Language Model (e.g., GPT-4o) is used to parse the input and extract structured editing parameters, including: (1) Action: The desired editing operation (e.g., “Add a mask”). (2) Method: The specific technique for obfuscation (e.g., “Gaussian blur,” “mosaic”). (3) Strength: The intensity level of the applied method. (4) Target: The specific object or region to be edited (e.g., “The first person on the left”). (5) Privacy List: A list of sensitive objects that require protection (e.g., “ID card,” “face”).

Automatic Segmentation: The Automatic Segmentation Module uses Grounded SAM to automatically detect and segment sensitive objects in the image. This module combines GroundingDINO

for object detection and SAM for segmentation, generating precise bounding boxes and masks based on the user’s natural language instructions. The process includes: (1) Object Detection: Identifying all objects in the image. (2) Segmentation Mask Generation: Creating masks for the identified sensitive regions (e.g., faces, license plates). (3) Soft Mask Creation[13]: Generating soft masks for smooth transitions between edited and non-edited areas.

Editing: The generated segmentation mask is then used for editing, applying the obfuscation method (Gaussian blur, mosaic, or generative masking) as specified in the user’s instruction. Gaussian blur and mosaic effects are applied using standard image processing methods. Generative editing is performed using the ACE++ framework, which allows for fine-grained, context-aware modifications.

Multi-Round Iterative Feedback: After the initial result is generated, the user can review the edited image and submit follow-up instructions (e.g., “increase the blur on the front face”). The system takes both the updated image and the original image as inputs, and re-executes the segmentation, editing, and fusion processes. This iterative mechanism ensures flexibility, user control, and reversibility throughout the editing process.

3 Conclusion

We introduce **PrivEdit**, a zero-shot, interactive image privacy editing system designed for automated sensitive information desensitization. Our approach demonstrates the system’s effectiveness in privacy protection and highlights its potential applications across various scenarios. Future research will focus on enhancing the system’s intelligence, improving its ability to understand complex instructions and edit quality, and expanding its capability to protect more types of sensitive information.

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